

ABRAHAM OGENAKOHGIE ANAVHEOBA

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RESEARCH INTERESTS

Bio-based heterogeneous catalysis; biodiesel production; renewable and sustainable fuels; waste-to-value materials; reaction engineering; process modelling and optimization; sustainable chemical process systems; artificial intelligence and machine learning applications in chemical engineering.

EDUCATION

University of Benin, Benin City, Nigeria

B.Eng. Chemical Engineering

September 2023

CGPA: 4.3/5.0

UNDERGRADUATE RESEARCH

Undergraduate Researcher

Department of Chemical Engineering, University of Benin, Benin City, Nigeria

2023

Research Project: *Synthesis of Bio-Based Heterogeneous Catalyst from Periwinkle Shells and Welders' Waste for Biodiesel Production Using Neem Seed Oil: Modelling and Optimization Studies*

Supervisor: Engr. Prof. K. O. Obahiagbon

- Conducted undergraduate research on the synthesis of bio-based heterogeneous catalysts from periwinkle shells and welders' waste for biodiesel production.
- Investigated neem seed oil as a renewable feedstock for sustainable biodiesel production.
- Studied catalyst performance and process variables affecting biodiesel production efficiency.
- Applied modelling and optimization approaches to evaluate biodiesel production conditions.

- Prepared and submitted a B.Eng. thesis under faculty supervision.
 - Developed research experience in sustainable fuels, catalyst development, and waste-to-value chemical engineering systems.
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INDUSTRIAL CHEMICAL ENGINEERING EXPERIENCE

Chemical Engineer

PZ Wilmar Nigeria Limited, Lagos, Nigeria

2024–2025

- Worked in edible oil refinery operations involving the processing of crude palm oil into refined edible oil products and palm oil fractions for major consumer brands including Mamador and Devon King's.
- Gained hands-on exposure to refining, fractionation, separation, process monitoring, and quality control within a large-scale edible oil refinery.
- Developed practical understanding of refinery process flow, plant operations, industrial safety, and food-grade manufacturing standards.
- Strengthened applied knowledge of chemical engineering principles in industrial oil processing and process operations.

Industrial Training Intern

Reliance Chemical Products Limited, Agbara, Ogun State, Nigeria

January 2022 – May 2022

- Completed industrial training in a sulphonation plant involved in LABSA-related chemical manufacturing.
 - Gained exposure to sulphonation operations, plant workflow, utilities, process monitoring, and industrial safety practices.
 - Observed the application of chemical engineering principles in large-scale production of sulphonic-acid-based products.
 - Developed foundational industrial experience in chemical process operations and manufacturing systems.
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TEACHING AND MENTORING EXPERIENCE

Peer Tutor / Undergraduate Mentor

Department of Chemical Engineering, University of Benin, Benin City, Nigeria

During undergraduate study

- Provided academic support to undergraduate students in core chemical engineering courses.

- Assisted junior students with tutorials, concept explanation, assignments, and technical problem-solving.
 - Supported students in understanding engineering calculations, process concepts, and course-related problem-solving methods.
 - Strengthened technical communication, mentoring, and academic support skills through regular peer teaching.
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COMPUTATIONAL, AI, AND SOFTWARE ENGINEERING EXPERIENCE

Backend Engineer / Software Engineer

2023–Present

- Built backend and distributed systems with emphasis on correctness, reliability, testing, and analytical problem-solving.
 - Developed strong programming and systems-thinking skills applicable to process modelling, simulation workflows, optimization, data analysis, and research software development.
 - Trained and evaluated AI/machine learning models for applied computational tasks.
 - Gained experience in data preprocessing, model training workflows, model evaluation, and iterative experimentation.
 - Contributed to open-source engineering and systems software projects involving large-scale codebases, protocol tooling, and automated testing.
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SELECTED PROFESSIONAL SOFTWARE EXPERIENCE

Senior Backend Engineer — Rust

AssetraExchange

January 2026 – Present

Software Engineer — Contract

GitStart / WebKit

August 2025 – December 2025

Backend Engineer — Contract

Isco Tech

May 2025 – July 2025

Backend Engineer — Contract

MathPrix

February 2025 – April 2025

Backend Engineer

CalendX

September 2023 – December 2024

OPEN-SOURCE CONTRIBUTIONS

- **WebKit:** Contributed to WebDriver BiDi implementation in a large-scale browser engine codebase.
 - **can-viewer-rs:** Implemented major DBC editor and serializer functionality in Rust.
 - **rusty-caravel:** Contributed CAN message parser improvements and tests.
 - **rsomeip:** Wrote end-to-end integration tests for SOME/IP protocol implementation.
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TECHNICAL SKILLS

Chemical Engineering: Heterogeneous catalysis, biodiesel production, edible oil refining, palm oil fractionation, sulphonation/LABSA process exposure, process operations, process safety, process modelling, process optimization.

Process Simulation and Modelling: Aspen HYSYS, process modelling, process optimization, simulation-based analysis.

Design and Engineering Tools: AutoCAD, SolidWorks.

Programming and Computation: Python, Rust, C++, JavaScript, SQL, computational workflows, data processing, backend systems, distributed systems.

Artificial Intelligence and Machine Learning: AI model training, machine learning workflows, data preprocessing, model evaluation, model optimization, experimental iteration, applied AI for data-driven problem-solving.

Software and Development Tools: Git, Docker.

PROFESSIONAL LINKS

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